



Finding the basis of a null space

Asked 12 years, 1 month ago Modified 6 years, 4 months ago Viewed 221k times



I am trying to understand why the method used in my linear algebra textbook to find the basis of the null space works. The textbook is 'Elementary Linear Algebra' by Anton.

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According to the textbook, the basis of the null space for the following matrix:



$$A = \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{pmatrix}$$

is found by first finding the reduced row echelon form, which leads to the following:

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (-3r - 4s - 2t, r, -2s, s, t, 0)$$

or, alternatively as

$$(x_1, x_2, x_3, x_4, x_5, x_6) = r(-3, 1, 0, 0, 0, 0) + s(-4, 0, -2, 1, 0, 0) + t(-2, 0, 0, 0, 1, 0)$$

This shows that the vectors

$$\mathbf{v}_1 = (-3, 1, 0, 0, 0, 0), \quad \mathbf{v}_2 = (-4, 0, -2, 1, 0, 0), \quad \mathbf{v}_3 = (-2, 0, 0, 0, 1, 0)$$

span the solution space.

It can be shown that for a homogenous linear system, this method always produces a basis for the solution space of the system.

Question

1. I don't understand why this method will always produce a basis for $Ax = 0$. Could someone please explain to me why this method will always work? If it helps to explain, I already understand the process of finding the basis of a column space and row space. I also understand why elementary row operations do not alter the null space of a matrix.
2. What specific properties of matrices or vector space that I need to be aware of in order to understand why this method works?

[linear-algebra](#)

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asked Dec 4, 2011 at 17:49



Sara

1,229

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4 Answers

Sorted by: Highest score (default)



58



The null space of A is the set of solutions to $A\mathbf{x} = \mathbf{0}$. To find this, you may take the augmented matrix $[A \mid \mathbf{0}]$ and row reduce to an echelon form. Note that every entry in the rightmost column of this matrix will always be 0 in the row reduction steps. So, we may as well just row reduce A , and when finding solutions to $A\mathbf{x} = \mathbf{0}$, just keep in mind that the missing column is all 0's.

Suppose after doing this, you obtain

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Now, look at the columns that do not contain any of the leading row entries. These columns correspond to the free variables of the solution set to $A\mathbf{x} = \mathbf{0}$. Note that at this point, we know the dimension of the null space is 3, since there are three free variables. That the null space has dimension 3 (and thus the solution set to $A\mathbf{x} = \mathbf{0}$ has three free variables) could have also been obtained by knowing that the dimension of the column space is 2 from the rank-nullity theorem.

The "free columns" in question are 2, 4, and 5. We may assign any value to their corresponding variable.

So, we set $x_2 = a$, $x_4 = b$, and $x_5 = c$, where a , b , and c are arbitrary.

Now solve for x_1 and x_3 :

The second row tells us $x_3 = -x_4 = -b$ and the first row tells us $x_1 = x_5 = c$.

So, the general solution to $A\mathbf{x} = \mathbf{0}$ is

$$\mathbf{x} = \begin{bmatrix} c \\ a \\ -b \\ b \\ c \end{bmatrix}$$

Let's pause for a second. We know:

- 1) The null space of A consists of all vectors of the form $\mathbf{x}$ above.
- 2) The dimension of the null space is 3.
- 3) We need three independent vectors for our basis for the null space.

So what we can do is take $\mathbf{x}$ and split it up as follows:

$$\mathbf{x} = \begin{bmatrix} c \\ a \\ -b \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} c \\ 0 \\ 0 \\ 0 \\ c \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -b \\ b \\ 0 \end{bmatrix}$$

$$= a \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

Each of the column vectors above are in the null space of A . Moreover, they are independent. Thus, they form a basis.

I'm not sure that this answers your question. I did a bit of "hand waving" here. What I glossed over were the facts:

1) The columns of the echelon form of A that do not contain leading row entries correspond to the "free variables" to $A\mathbf{x} = \mathbf{0}$. If the number of these columns is r , then the dimension of the null space is r (again, if you know the dimension of the column space, you can see that the dimension of the null space must be the number of these columns from the rank-nullity theorem).

2) If you split up the general solution to $A\mathbf{x} = \mathbf{0}$ as done above, then these vectors will be independent (and span of course since you'll have r of them).

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edited Feb 11, 2016 at 10:55

answered Dec 4, 2011 at 18:37



David Mitra

74.4k

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▲ :Thank you for explaining the procedure of finding the basis of the null space. I find it to be much clearer than how the textbook explained it. However, I am still not clear why splitting the general solution will produce a set of vectors that will span the null space. Also, the set of vectors are independent because splitting $\mathbf{x}$ will always produce non-zero vectors, hence $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n$ will have the trivial solution only. Is this right? – Sara Dec 5, 2011 at 16:31 ✎

1 ▲ @sara Sorry for the last comment. When you split $\mathbf{x}$ up, you split it into vectors with disjoint supports, so you'll have independence. For spanning: Splitting the general solution up as above will give the necessary number of vectors for a basis of the null space, since you decompose $\mathbf{x}$ into a linear combination of as many vectors as you have free variables (and the number of free variables is the number of "free columns", which in turn is the dimension of the null space by the rank-nullity theorem). – David Mitra Dec 5, 2011 at 18:34

▲ Thank you very much for your clarification – Sara Dec 9, 2011 at 16:49 ✎

1 ▲ @AkashPatalwanshi For any i , at most one of the vectors has a nonzero i 'th coordinate. – David Mitra May 9, 2022 at 5:57

1 ▲ @AkashPatalwanshi Yes, those vectors are not disjointly supported; so verifying their independence takes a bit more work. – David Mitra May 9, 2022 at 15:03 ✎



There is a mechanical part in the accepted (+1) answer that may possibly need a more step-by-step explanation, namely the process behind

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what we can do is take $\mathbf{x}$ and split it up



Reducing the matrix A to the reduced row echelon (rref) form results in:



$$A = \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 0 & 4 & 2 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

There are three pivot columns corresponding to the pivot 1's in red, and the rank-nullity theorem tells us that there are $n - r = 3$ vectors in any basis of the $N(A)$, corresponding to the free variables.

At this point it is worth remembering where all this comes from: the homogeneous system of linear equations $A\vec{x} = \vec{0}$, which has now been reduced to:

$$1x_1 + 3x_2 + 0x_3 + 4x_4 + 2x_5 + 0x_6 = 0$$

$$0x_1 + 0x_2 + 1x_3 + 2x_4 + 0x_5 + 0x_6 = 0$$

$$0x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 + 1x_6 = 0$$

$$0x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 + 0x_6 = 0$$

Expressing the pivot variables in terms of the free variables:

$$1x_1 = -3x_2 - 4x_4 - 2x_5$$

$$1x_3 = -2x_4$$

$$1x_6 = 0$$

immediately shows the way the basis vectors of the $N(A)$ will be filled in from their "skeleton" form simply indicating the column where the free variable in question is located (i.e. x_2, x_4, x_5):

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\} \rightarrow x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

to the final form:

$$x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \rightarrow \text{basis } N(A) = \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

So it boils down to changing the signs of the entries in the rref, and keeping track of the free variables.

```
require("pracma")
A = matrix(c(1,3,-2,0,2,0,2,6,-5,-2,4,-3,0,0,5,10,0,15,2,6,0,8,4,18), nrow=4, byrow=T)
x2= c(-3,1,0,0,0,0); A %**% x2; x4=c(-4,0,-2,1,0,0); A %**% x4; x5=c(-2,0,0,0,1,0); A %**% x5
```

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answered Sep 1, 2017 at 12:52



Antoni Parellada

8,779 5 43 134

Solution

1

The matrix $C_{\rho}^{m \times n}$ is *map* between C^n and C^m .

Consider the vectors $x \in C^n, y \in C^m$. The matrix A maps C^n to C^m ; A^* maps C^m to C^n .

The Fundamental Theorem of Linear Algebra appears in the guise of a map.

$$\begin{aligned} Ax &= y & C^n &\mapsto C^m \\ A^*y &= x & C^m &\mapsto C^n \end{aligned}$$

I already understand the process of finding the basis of a column space and row space

Since the machinations are clear, we choose a simpler matrix which caters to mental manipulation:

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & 6 & -5 \end{bmatrix} \in C_2^{3 \times 2}$$

The only nontrivial null space is $\mathcal{N}(A)$.

The augmented matrix is

$$\begin{bmatrix} A^* & I_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 3 & 6 & 0 & 1 & 0 \\ -2 & -5 & 0 & 0 & 1 \end{bmatrix}$$

You perform a sequence of elementary row operations to reduce the left-hand component to row echelon form. Call the first operation which clears the first column E_1 .

$$E_1 \begin{bmatrix} A^* & I_3 \end{bmatrix} = \begin{bmatrix} E_1 A^* & E_1 I_3 \end{bmatrix}$$

The second step clears the second column:

$$\mathbf{E}_2 \begin{bmatrix} \mathbf{E}_1 \mathbf{A}^* & \mathbf{E}_1 \mathbf{I}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}^* & \mathbf{E}_2 \mathbf{E}_1 \mathbf{I}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & -3 & 1 & 0 \\ 0 & -1 & 2 & 0 & 1 \end{bmatrix}$$

Your question is how the null space vector appeared:

$$\begin{bmatrix} 1 & 3 & -2 \\ 2 & 6 & -5 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The answer involves a change of perspective.

The elementary operations are sculpted to sweep through the matrix and clear successive rows. This much is clear. The perspective change involves looking at the augmented matrix as collection of row vectors:

$$\begin{bmatrix} \mathbf{A}^* & \mathbf{I}_3 \end{bmatrix} = \begin{bmatrix} u_1^* & v_1^* \\ u_2^* & v_2^* \\ u_3^* & v_3^* \end{bmatrix}$$

By construction, the u vectors in $\mathbb{C}^3$ are linked to the v vectors in $\mathbb{C}^2$ via

$$v_k^* \mathbf{A}^* = u_k^*, \quad k = 1, 2, 3$$

The final form we seek is given by

$$\begin{aligned} (v_k^* \mathbf{A}^*)^* &= (u_k^*)^* \\ \mathbf{A} v_k &= u_k \end{aligned}$$

The mapping action of the matrix $\mathbf{A}$ defines the augmented matrix. When manipulation creates a $u = \mathbf{0}$ vector, the corresponding v vector must be in the null space.

Colloquial mathematics

The augmented matrix is a tool to study the mapping action of a matrix between the vector spaces $\mathbb{C}^m$ and $\mathbb{C}^n$. To find null space vectors, manipulate the left-hand side to create a zero row. A null space vector appears as a row vector on the right-hand side.

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answered May 6, 2017 at 23:32



dantopa

10.3k

10

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82



The software Mathematica can find a null-space spanning set for Matrices given with exact coefficients. It can find exactly the answer your book gives:

-3

```
NullSpace[{{1, 3, -2, 0, 2, 0}, {2, 6, -5, -2, 4, -3}, {0, 0, 5, 10, 0, 15}, {2, 6, 0, 8, 4, 18}}]
```



gives

 $\{-2, 0, 0, 0, 1, 0\}, \{-4, 0, -2, 1, 0, 0\}, \{-3, 1, 0, 0, 0, 0\}$ 

I suggest you [start here](#) if you are interested in how this is done in a computationally efficient and of course provably correct way. That Wikipedia section for instance points out, that in practice the so called Bareiss algorithm is more useful than Gaussian elimination.

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answered Oct 4, 2016 at 15:58



masterxilo

376 2 9



(My apologies: this is a different question. However, my original complaint stands: this post did not need to be bumped to the main page after nearly 5 years.) – [Eric Stucky](#) Oct 4, 2016 at 16:05